

eduTracker Enterprise ERP - Full Business Documentation

1. Executive Summary

eduTracker is a comprehensive, on-premise Enterprise Resource Planning (ERP) system designed specifically for educational institutions (K-12 schools, colleges, and academies). It digitizes and automates the entire lifecycle of school management—from the moment a student inquires about admission, through their daily academics and fee payments, to staff payroll and final graduation certificates.

2. Core Modules & Business Logic

2.1. Student Information System (SIS) & Admissions

- **Admissions Inquiry:** Tracks prospective students, follow-up dates, and conversion rates. Allows the school to measure marketing success.
- **Student Enrollment:** Digitizes the admission process. Captures vital demographics, medical info, previous academic history, and parent/guardian contact details.
- **Document Management:** Automatically generates ID Cards, Leaving Certificates, and Character Certificates using customizable templates based on the student's profile.

2.2. Academics & Timetable Management

- **Class & Section Management:** Organizes the school hierarchy (e.g., Grade 10, Section A).
- **Subject & Syllabus Mapping:** Assigns specific subjects to classes and tracks syllabus progression.
- **Timetable (Periods):** Manages the daily schedule, preventing teacher overlap and ensuring optimal resource allocation.

2.3. Attendance Tracking

- **Student Attendance:** Daily tracking of student presence (Present, Absent, Late, Excused). Feeds directly into the analytics dashboard and report cards.
- **Staff/HR Attendance:** Tracks teacher and staff attendance, which is automatically linked to the Payroll module for accurate salary calculations.

2.4. Examination & Grading Engine

- **Exam Types:** Configure custom exams (Mid-terms, Finals, Weekly Quizzes) with base marks.
- **Marks Entry & Locking:** Teachers enter marks through a secure portal. Once verified, marks are "Locked" to prevent unauthorized tampering.

- **Academic Report Generation:** Automatically compiles marks, attendance, and teacher remarks into a final Report Card.

2.5. Question Bank & Automated Paper Generator

- **Question Bank:** A centralized repository where teachers can store questions categorized by subject, difficulty, and class.
- **Question Paper Generator:** Teachers can dynamically generate formatted exam papers by selecting questions from the bank or using pre-defined templates, saving hours of formatting time.

2.6. Human Resources (HR) & Payroll

- **Staff Management:** Maintains records for Teachers, Drivers, Librarians, Principals, and Admins.
- **Leave Management:** Staff can apply for leave through the portal. Admins can approve/reject, automatically adjusting payroll.
- **Payroll Processing:** Automatically calculates monthly salaries based on base pay, attendance deductions, and approved leaves. Generates digital payslips.

2.7. Finance & Fee Management

- **Fee Structure:** Defines what different classes owe (Tuition, Transport, Library fees).
- **Fee Collection:** Tracks payments, generates automated invoices, and issues digital receipts to parents.
- **Outstanding Dues:** Provides the Principal with a dashboard of unpaid fees to easily track the school's revenue health.

2.8. Operations Management

- **Library Management:** Tracks books, issues/returns to students and staff, and automatically calculates late fines.
- **Transport Management:** Manages the school bus fleet, assigns students to specific routes, maps driver assignments, and integrates transport fees into the student's billing profile.
- **Inventory Management:** Tracks physical school assets (chalk, lab equipment, computers, uniforms). Monitors stock levels to prevent shortages.

2.9. Artificial Intelligence (AI) Insights

- **Performance Analytics:** Analyzes historical marks and attendance to identify struggling students before they fail.
- **Automated Summaries:** Generates human-readable, AI-driven insights on a student's term performance to assist teachers in writing report card remarks.

2.10. Security, Audit, & System Settings

- **Role-Based Access Control (RBAC):** Strict permissions ensure Teachers only see their classes, HR only sees payroll, and the Principal sees everything.
 - **Audit Logging:** Tracks every critical action (e.g., "User X changed Student Y's marks on Date Z"). Essential for institutional accountability.
 - **White-labeling:** The software can be dynamically branded with the specific school's Logo, Name, and Colors.
 - **Enterprise Licensing:** The system is protected by a cryptographic JWT license key that automatically locks the software if the school's subscription expires.
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3. The Business Value (Why Schools Buy This)

When selling eduTracker to a Principal or School Board, focus on these three pillars:

1. **Revenue Protection:** Automated fee tracking, late fine calculations (library), and transport billing ensure the school collects 100% of what it is owed.
2. **Time Savings:** Automated Report Cards, automated Payroll, and the Question Paper Generator save teachers and administrators thousands of hours per year.
3. **Data Security & Ownership:** Because it is an *On-Premise* solution, the school has ultimate peace of mind. Their sensitive student data never leaves their physical building, distinguishing eduTracker from cloud-based competitors.